

Synthetic data

Enhancing open and reproducible social science while protecting participants' privacy

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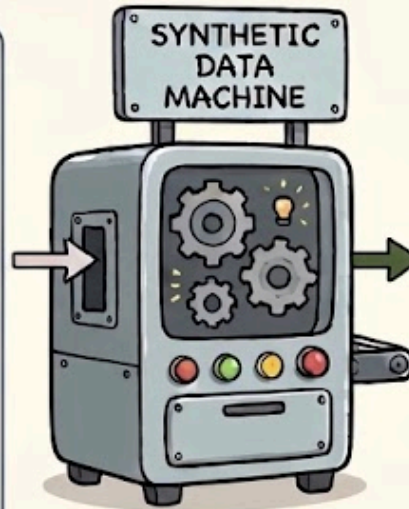
Utrecht University & Statistics Netherlands

Synthetic data



Observed Data

age	height	weight	testicular volume
5.78	124.8	25.1	NA
5.82	NA	NA	NA
7.04	130.5	27.0	NA
8.87	145.0	38.2	2
9.20	125.8	22.0	3
6.02	128.1	27.8	NA
7.39	135.9	45.3	NA
8.91	137.8	30.0	2



Synthetic Data

age	height	weight	testicular volume
5.92	128.5	NA	1
5.65	122.0	23.9	1
7.21	NA	28.1	1
9.12	149.1	40.5	NA
9.45	121.7	21.0	2
6.05	130.2	29.5	1
7.41	136.5	44.8	1
8.93	138.2	31.0	2



Is anonymizing not enough?

If done well, anonymization might suffice.

But: data might be linked in surprising ways.

With enough variables, every record becomes unique.

Netflix prize data

Dataset with movie ratings provided by Netflix: movie rating (1-5); submission date

Narayanan & Shmatikov (2008) showed that 99% of the records can be identified based on

- 8 movie ratings of which 2 were allowed to be incorrect, and dates within a 2 weeks margin.
- Movie ratings can be obtained from external data sources

Synthetic data generation pipeline

1. Privacy risk assessment: *anonymization; identifying sensitive variables*
2. What's the goal of the synthetic data: *Dummy data, teaching, replication, novel research?*
3. Choosing a synthetic data model *that fits this goal*
4. Evaluating risk and utility
5. Refinements necessary?

Generating synthetic data

Capture the most important information about the data in a *generative* model

- Parameters are learned from real data; as opposed to simulated data
- Information that flows into the model can be bounded
- Relationships in real data can be modelled explicitly

Draw new samples from this model

Don't you ever!

What's on the agenda today?

Can you generate a synthetic version of the following dataset:

"Thom Volker",29,"M","Utrecht","The Netherlands",183,71,43

"Jolanda Neff",33,"F","Sankt Gallen","Switzerland",168,53,38

"Roger Federer",44,"M","Basel","Switzerland",185,85,46

...

+

↑

Conditional modelling

One (regression) model per variable

- Offers a lot of flexibility
- Allows easy model checking (posterior predictive checks)
- Easier to make refinements
- Transparent what information ends up in synthetic data

Software: `synthpop`¹ (R, python); `simPop` (R)²

(Non-parametric) joint modelling

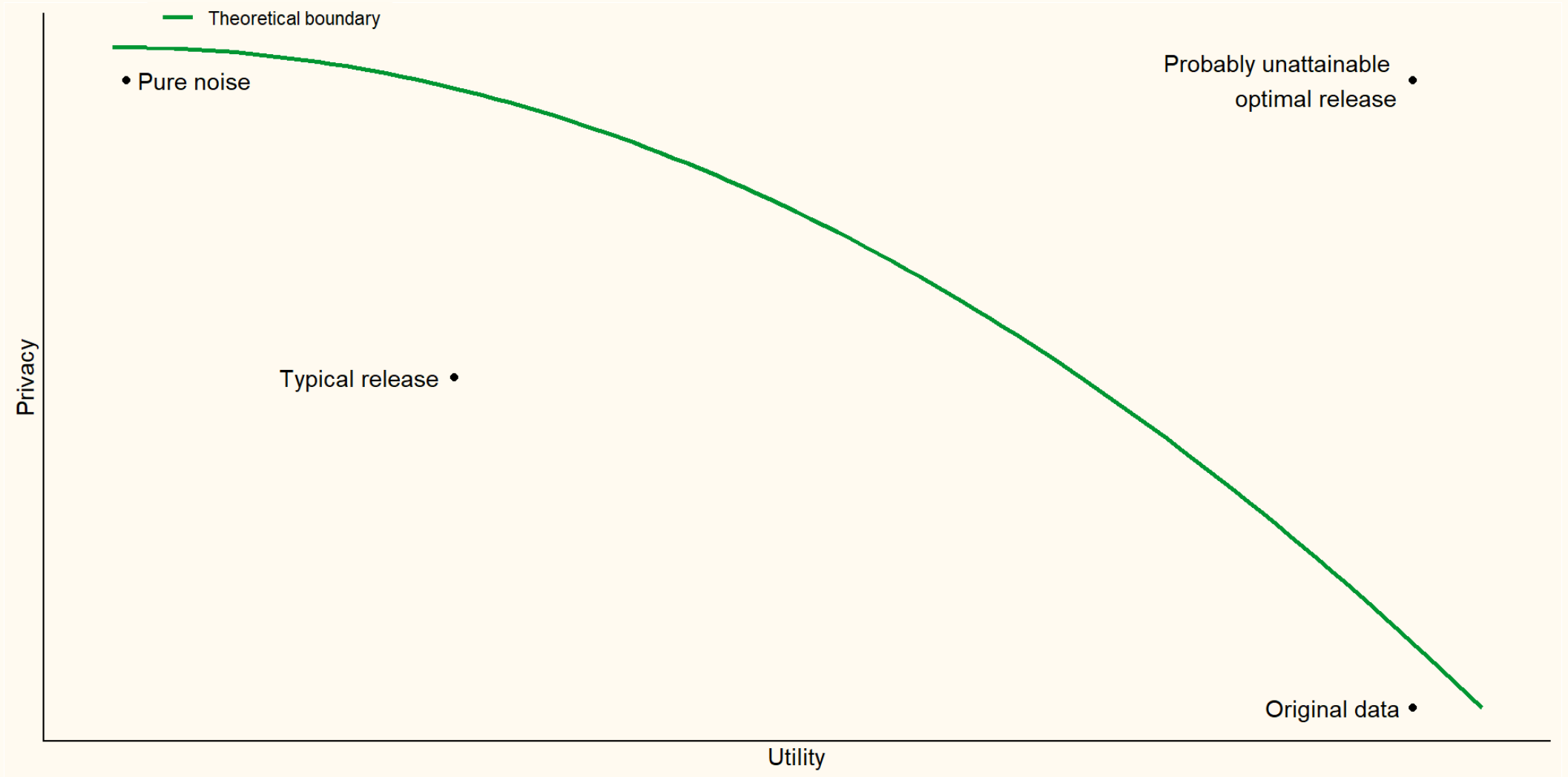
Often based on deep learning

Superior for unstructured data (images, geo-spatial data, fMRI)

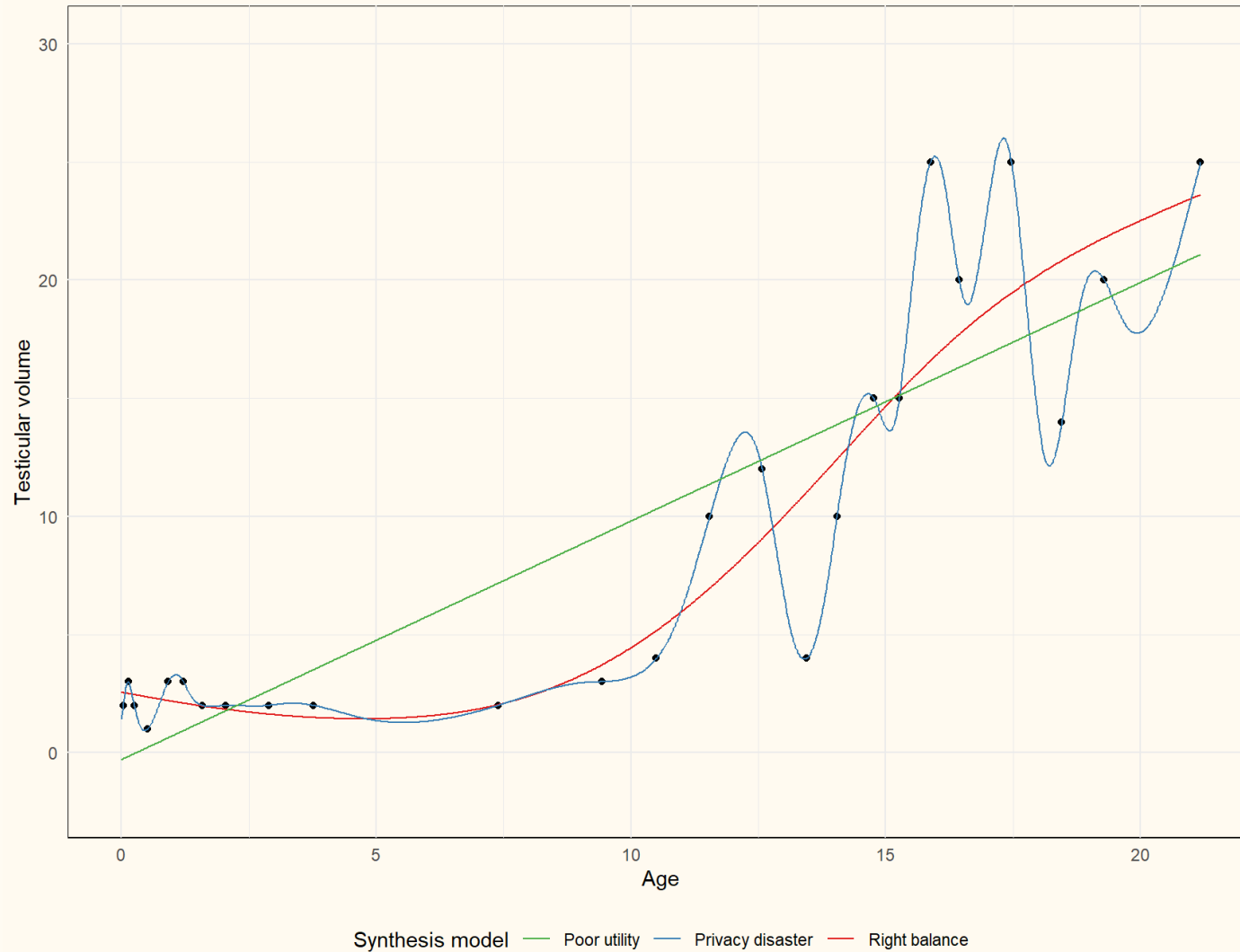
Hard to know and communicate what information ends up in the synthetic data

Software: Synthetic Data Vault (sdv)¹, synthcity²; RGAN³.

The privacy-utility trade-off



The privacy-utility trade-off illustrated



Bounding information flow into synthesis model

Traditionally: restricting to simpler models

- (Generalized) linear models; identifying influential cases (leverage; degrees of freedom per parameter)

For non-parametric models:

- Regularization
- Differential privacy

Differential privacy

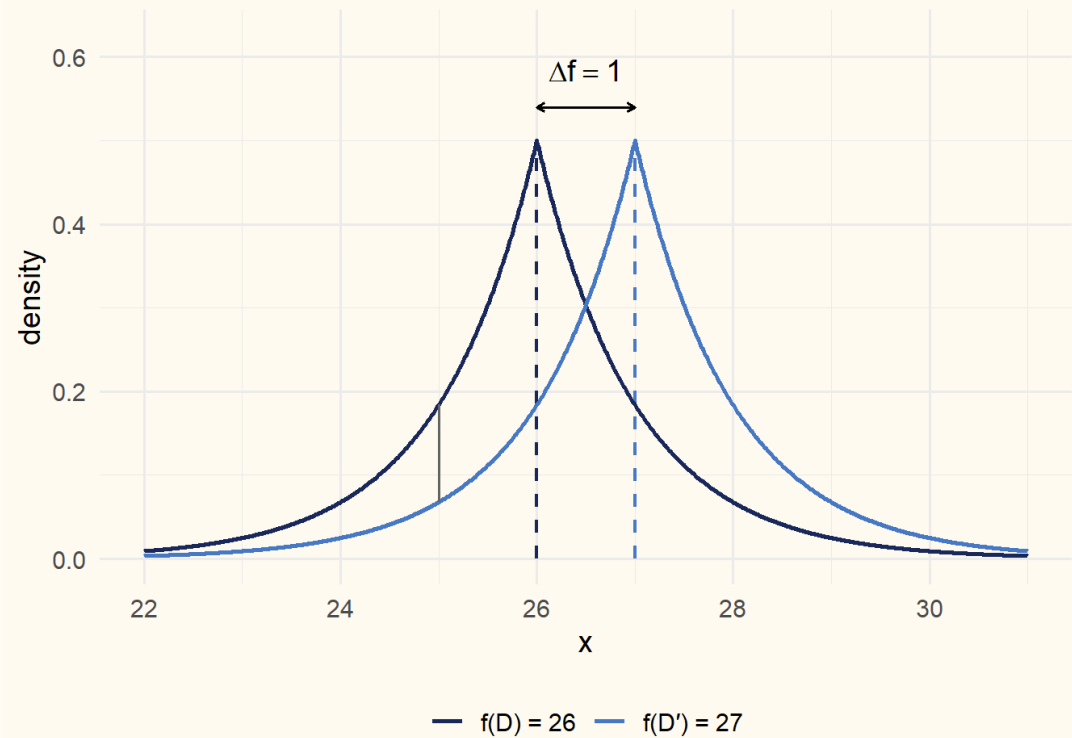
Only formal privacy framework as of yet

- Sensitivity: the maximum influence a single individual can have on a model
- Adds noise such that the results are almost equivalent regardless of whether this maximally influential individual was present in the data

Differential privacy for a count

Say we have $n = 100$ individuals in our data without or with a disease (o/I)

- Goal: estimate number of individuals with disease
- **Sensitivity:** Altering one individual affects our estimate by at most



Advantages of synthetic data

Privacy can be well protected

- No real records anymore
- Information that goes into synthesis model can be bounded

A large amount of information can be preserved

- Important information can be modelled explicitly

Disadvantages

Quality of synthetic data depends entirely on synthesis model

- Too simple models might omit important relationships
- Complex synthesis models obscure what information is reproduced in the synthetic data

Some helpful resources

- Overview: Drechsler & Haensch (2024). 30 years of synthetic data. *Statistical Science*, 39(2), 221-242.
- Tutorial: Volker (2025). Generating and evaluating synthetic data in R. LMU Open Science Center.
<https://doi.org/10.5281/zenodo.18318778>
- **synthpop** webpage with practical advice:
<https://www.synthpop.org.uk/resources.html>

Questions / comments?

Feel free to reach out: t.b.volker@uu.nl

Slides: <https://github.com/thomvolker/dgp-synthetic/>